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Direct, non-destructive quantitative measurement of an active pharmaceutical ingredient in an intact capsule formulation using Raman spectroscopy

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Abstract

The active pharmaceutical ingredient (ambroxol) in an intact capsule formulation has been non-destructively quantified using Raman spectroscopy. To improve the problem of insufficient representive sampling inherent in Raman measurements, we have employed a wide area illumination (WAI) scheme that enables much improved sample coverage through a circular excitation laser spot with a 6 mm diameter. One of the anticipated sources of variation for this measurement was variation in the capsule shells. However, the WAI scheme significantly decreased the spectral variation among empty capsules compared to a measurement with a traditional small-spot excitation. Therefore, measurement variations emanating from the capsule shell did not significantly influence the accuracy of the determination of ambroxol concentrations. The resulting standard error of prediction (SEP) using the WAI scheme was comparable to that from previous Raman measurements which used a conventional small-spot excitation and employed a sampling scheme that involved rotation of an ambroxol pellet. It is further noteworthy that the SEP was also similar to that obtained from the use of transmission NIR spectroscopy, which was achieved by collection of spectra of the powdered capsule contents removed from the shell. The proposed Raman measurement using the WAI scheme in this case was sufficient to achieve the quantitative measurement of the active pharmaceutical ingredient (API) content of capsules non-destructively.
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Keywords: Pharmaceutical capsule; Ambroxol; Non-destructive analysis; Raman spectroscopy; Wide area illumination

1. Introduction

The direct measurement of the active pharmaceutical ingredient (API) in diverse formulations without the need for sample preparation has always been of great interest to those in pharmaceutical quality control operations. The most obvious benefits of such an approach include not only the realization of time savings and the ability to avoid the use (and disposal) of solvents, but also the potential for the facile automation of such analyses. Included with this is the potential to carry this automation to the production line. Such opportunities have become increasingly attractive since the process analytical technology (PAT) guidance was issued by the U.S. Food & Drug Administration (FDA) in 2004 [1,2].

One of the most challenging analyses to accomplish without sample manipulation is the direct measurement of an API in a formulation within an intact capsule. Sample handling of such a formulation often involves tedious procedures that entail quantitative transfer of the capsule contents from the shell to a preparation vessel. Because the shells are typically composed chiefly of polymers, dissolving the capsules to extract the contents, though simpler than a manual transfer, is often not practical. Thus, the quality assurance assay of capsules can be greatly simplified if sample preparation can be obviated.

Previously, Niemczyk et al. [3] attempted to determine the bucindolol concentration in intact capsules using Raman spec-

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troscopy. The results of their study suggested this approach was feasible. However, the largest source of error originated from sample inhomogeneity. This, in turn, stemmed from inaccurate compositional representation of their samples due to the small laser illumination area.

To circumvent this drawback inherent in standard Raman measurements, we have previously demonstrated a novel Raman collection system that makes use of a wide area illumination (WAI) scheme [4,5]. The basis for this approach is the application of a circular illumination spot that is 6 mm in diameter (area: 28.3 mm^2). We previously evaluated the performance of this scheme to determine the naproxen concentration in tablets [4]. Raman spectra collected by the WAI scheme were much more reproducible and better representative of the correct API composition of the tablets compared to a traditional small-spot scheme.

The purpose of this paper is to demonstrate the utility of the WAI scheme for the compositional analysis of an API (ambroxol) in a formulation directly through a capsule shell. With the larger coverage area of the excitation laser, the measurement accuracy can be improved by better representative sampling. Raman spectra of capsules containing different concentrations of ambroxol and lactose (excipient) were collected using the WAI scheme. It was shown that the spectral variation from one capsule shell to another was not significant. Therefore, capsule shell compositional and morphological variations did not negatively influence the calibration performance. Partial least squares (PLS) regression [6–8] was performed to model the ambroxol concentrations and the resulting prediction capability was shown to be reliable. The standard error of prediction (SEP) obtained in this study was comparable to that obtained in an earlier study in our laboratory in which the same formulations were pelletized prior to measurement [9]. However, the pelletized samples had to be rotated during the measurement process in order to achieve representative sampling using a conventional small-spot scheme (approximately 80 μm spot size).

In an attempt to ascertain the performance of this approach, we compared the results from this study with data obtained with near-Infrared (NIR) transmission spectra (pathlength: 2 mm) of ambroxol powder formulations. PLS models were also constructed from the NIR data in order to predict the API contents of the powdered formulations. The SEPs obtained for NIR transmission measurements of formulation powders were similar to those obtained for the Raman measurements of formulations in intact capsules using the WAI scheme.

2. Experimental

2.1. Sample preparation and spectral collection

Thirteen different samples (concentration range: 10.0–16.0 wt.% of ambroxol) were prepared by mixing appropriate amounts of ambroxol and lactose (both purchased from Sigma–Aldrich, Milwaukee, WI, USA). The mixed powder samples (430 mg) were transferred into a capsule shell. Fig. 1 shows the picture of the capsules used in this study with the corresponding dimensions. The bottom portion of the capsule

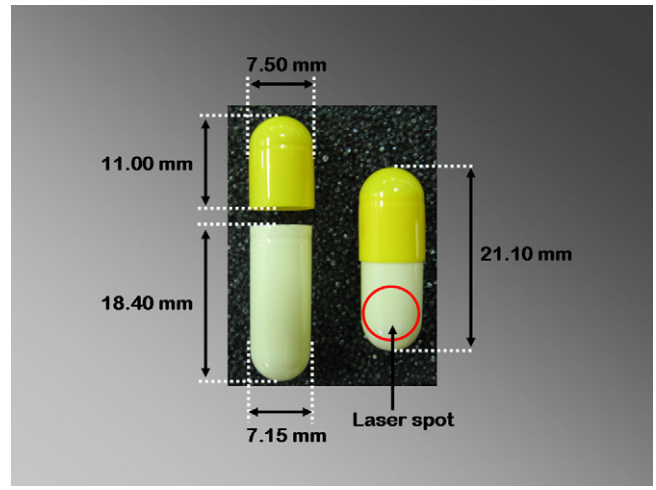

Fig. 1. The picture of a capsule and corresponding dimensions. Laser illumination was at the lower part of the capsule.

was illuminated for each measurement. The corresponding illumination spot is designated as a circle in the figure.

The schematic diagram of the wide area illumination (WAI) scheme is shown in Fig. 2 [4]. The excitation laser (785 nm, diode laser) was introduced and magnified to form a circular illumination area of 6 mm diameter (area: 28.3 mm^2) to cover a large sample area. The scattered radiation was then collected by an array of 50 optical fibers and delivered to the CCD detector. The radiation from the optical fibers was dispersed via a holographic transmission grating and a combined signal was integrated over the total area illuminated. Capsules were simply located at the focal point of the laser for measurement. To correct for the Raman intensity variation that could result from normal laser power fluctuations, isobutyric anhydride external standard, sealed in a 2 mm thick rectangular quartz cell, was positioned at 22 mm in front of the capsule. The resulting Raman spectrum was the summation of spectral features from both the sample (including the ambroxol powder and capsule shell) and the external standard. To utilize an external standard to correct Raman intensity, it is required to have at least one band of the standard that does not overlap the sample bands. Isobutyric anhydride is chosen because it has the band at 1810 cm^{-1} (carbonyl stretching) that seldom overlaps with bands of general pharmaceutical products. In addition, isobutyric anhydride is chemically and physically stable (b.p.: 182 °C) for use over a long period of time. This allowed the non-overlapping isobutyric anhydride peak (1810 cm^{-1}) to be used to correct the intensity variation

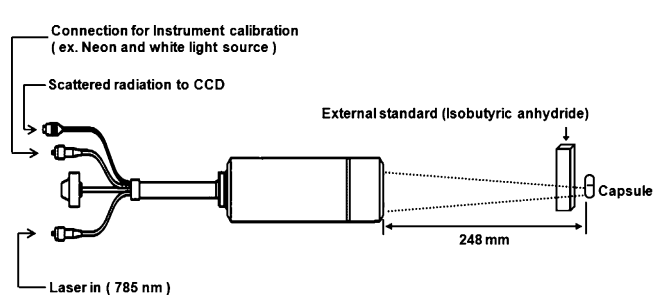

Fig. 2. Schematic diagram of the wide area illumination (WAI) scheme.

resulting from laser fluctuations. Raman spectra were collected using an exposure time of 3 s and 32 accumulations (resolution: 4 cm^{-1}). Triplicate spectra were collected for each sample.

All of the NIR spectra were collected using an ABB Bomem MB-160 bench-top FT-NIR spectrometer (Quebec City, Quebec, Canada) equipped with a tungsten-halogen source and an InAs detector. Powder samples were transferred into a 2 mm path-length cell (transmission) made of quartz prior to NIR spectral collection. Teflon (PTFE: polytetrafluoroethylene) powder was used as a background for all of the sample spectra. Each NIR spectrum was obtained by accumulating 64 scans at a resolution of 8 cm^{-1}. Triplicate spectra were collected for each sample.

For the construction of PLS (partial least squares) calibrations, the 13 samples were divided into eight samples (24 spectra, 10.0–16.0 wt.% range) for the calibration set and five samples (15 spectra, 11.0–15.0 wt.% range) for the validation set. All of the calculations including baseline-corrections, intensity corrections and PLS regressions were accomplished using Matlab version 7.0 (The Math-Works Inc., Natick, MA, USA).

3. Results and discussion

3.1. Raman spectral features

Fig. 3 shows individual Raman spectra of the ambroxol and lactose raw materials, an empty capsule shell, isobutyric anhydride external standard and a 12% ambroxol capsule sample. The spectral features of ambroxol and lactose are significantly different due to the substantial structural differences noted in the previous publication [9]. The Raman spectral features of the capsule shell are relatively simple. The manifestation of an underlying weak fluorescence background from the capsule material is likely due to the presence of the pigments that confer its yellowish ivory or pale golden rod appearance. The Raman spectrum of isobutyric anhydride, used as a synchronously collected external standard, shows a non-overlapping band at 1810 cm^{-1} (C=O stretch) that is available as a reference peak to compensate for laser power fluctuations. Overall, as expected, the visual inspection of the Raman spectrum of the 12% ambroxol capsule suggests that it corresponds to the additive spectral features conferred individually from ambroxol, lactose, the capsule shell and isobutyric anhydride.

For reliable quantitative analysis of the API in an intact capsule, the spectral variation contributed from the variance in the capsule shell should be minimal. To evaluate the extent of this variation, we have collected Raman spectra of eight empty capsules (three spectra per capsule: total of 24 spectra). These spectra were baseline-corrected at nine different positions (1873, 1724, 1618, 1216, 998, 807, 684, 271 and 189 cm^{-1}) to minimize the underlying fluorescence background. The resulting spectra were further normalized by dividing each by the area of the isobutyric anhydride band integrated from 1850 to 1730 cm^{-1}. The results are shown in Fig. 4. The top and bottom plots correspond to only the baseline-corrected spectra and normalized spectra, respectively. The various spectra are similar to each other in both the top and bottom plots. The normalized data appear to be more reproducible based on a simple visual examination of the spectral features and the enlarged isobutyric anhydride band.

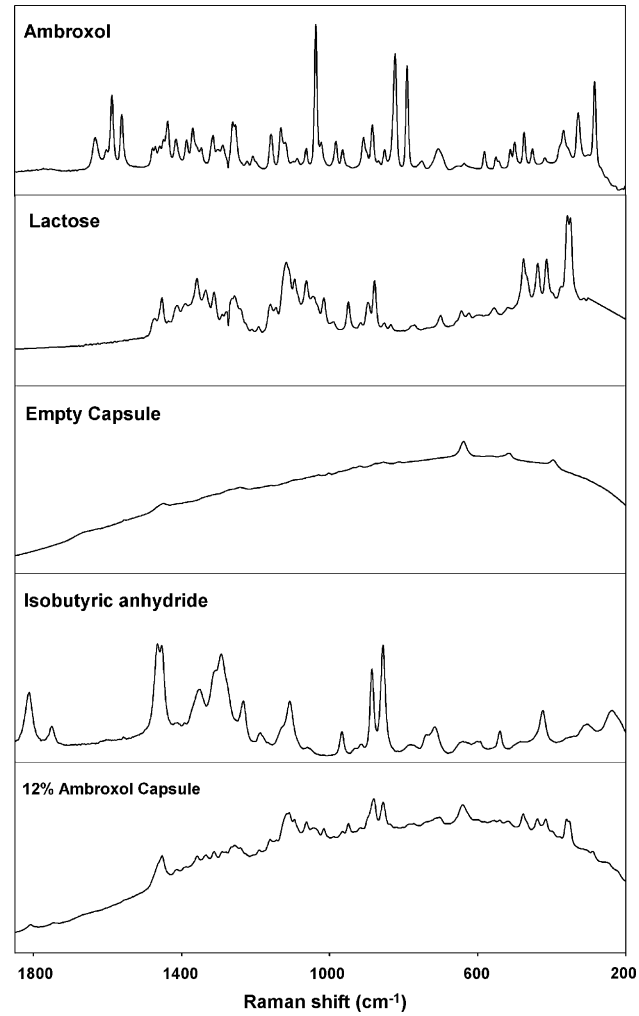

Fig. 3. Raman spectra of ambroxol, lactose, empty capsule, isobutyric anhydride and a 12% ambroxol capsule sample.

In order to characterize the influence of the spectral variation of the empty capsules on the calibration performance, the relative standard deviation (R.S.D.) of the capsule spectra at each wavenumber were calculated. The standard deviations were initially calculated using the spectra of the empty capsule shells, then divided by the corresponding average intensities of the spectra in the calibration set (capsules containing ambroxol). The resulting calculations were expressed as percentages. By doing this, the relative spectral variation contributed by the capsule shells themselves can be visualized. The results are shown in Fig. 4. The R.S.D.s are low and relatively constant throughout the spectral range except the 800–200 cm^{-1} range. The higher R.S.D.s in this range probably originates from the imperfect baseline-correction scheme that occurred due to the steeper, non-linear baseline characteristics of this region (see Fig. 3). This analysis suggests that the 800–200 cm^{-1} range should be excluded from the calibration model. Examination of the remaining spectral region (1700–800 cm^{-1}) indicates that there should be sufficient useful spectral information to create a robust model.

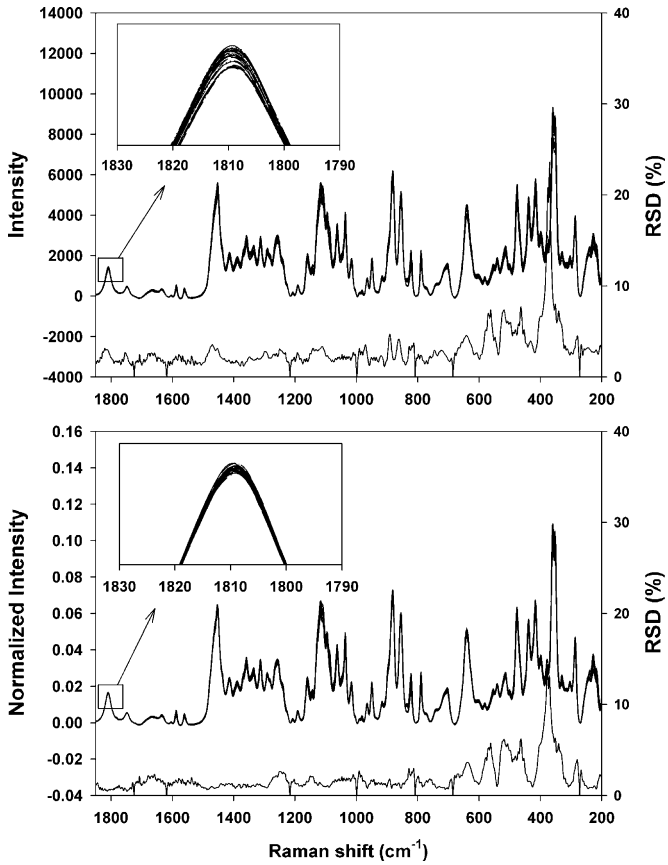

Fig. 4. Baseline-corrected spectra (top) and baseline-corrected and normalized spectra (bottom) of eight empty capsules (three spectra per capsule: total 24 spectra). The 1830–1790 cm^{-1} range is magnified in both plots. In addition, the calculated relative standard deviation (R.S.D.) at each wavenumber is display in each plot.

The average R.S.D.s of the baseline-corrected and normalized spectra in the 1700–800 cm^{-1} range were 2.6 and 1.4%, respectively. The R.S.D.s were improved by correcting Raman intensity using the area of the external standard peak. The R.S.D.s of normalized spectra in the 1850–1730 cm^{-1} range, where the isobutyric anhydride band of interest is located, are quite low. The spectral variation from the capsule shell in the final calibration range (1700–800 cm^{-1}) is not significant relative to the intensity of the spectral data collected from the ambroxol capsule. Therefore, this suggests that the feasibility of determining the ambroxol concentration directly through the capsule would not be significantly limited by interference from the capsule shell.

As discussed in the introduction, the major drawback of using Raman spectroscopy for the analysis of pharmaceutical solid samples is the inaccurate representation of the sample composition because of the small illumination area commensurate with a standard measurement scheme. This phenomenon can be observed in the case of the 12% ambroxol capsule sample. Fig. 5 shows a plot in which the 1240–1100 cm^{-1} range is expanded. The Raman spectra in this figure were collected at 5 different spots using the same capsule sample with a conventional illumination scheme having a spot size of approximately 80 μm (non-contact Raman probe with a 10 mm focal length).

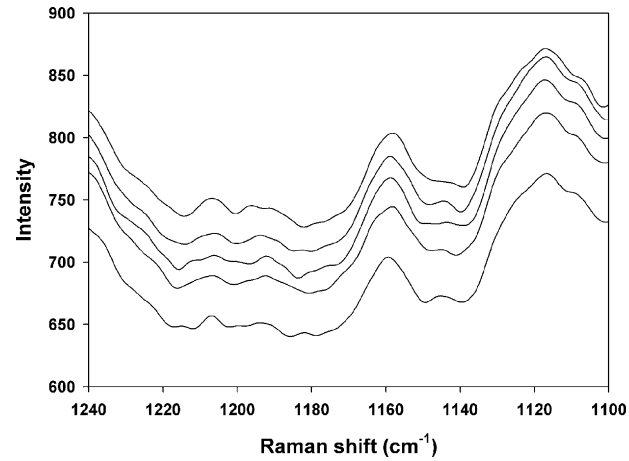

Fig. 5. Raman spectra of a 12% ambroxol sample (1240–1100 cm^{-1} range) collected at five different spots on the same capsule sample using a conventional illumination scheme with a spot size of 80 μm.

The resulting Raman spectra are slightly different from each other, even though the spectra were acquired from the same sample. None of these individual spectra satisfactorily represent the average composition of the ambroxol in the capsule. The use of the WAI scheme would be expected to eliminate or substantially alleviate the problem of representative sampling for pharmaceutical capsule analysis. This same principle was demonstrated in the measurement of naproxen tablets [4].

3.2. Partial least squares (PLS) regression using Raman spectral data

To determine ambroxol concentrations, the Raman spectra were initially baseline-corrected at nine different positions as in Fig. 4. Subsequently, the band area of isobutyric anhydride integrated from 1850 to 1730 cm^{-1} was calculated and each sample spectrum was divided by the calculated area to compensate for any possible Raman intensity variations that did not result from compositional variation. For PLS calibrations [6–8], the 1700–800 cm^{-1} range was used. The range of the degraded spectral reproducibility (800–200 cm^{-1} range) described earlier as being attributed to improper baseline-correction (Fig. 4) was excluded from the modeling procedure. However, for the purpose of comparison, we also constructed models using the whole spectral range (1700–200 cm^{-1} range).

The optimum number of factors was identified by examining the pattern of the decreasing standard error of cross-validation (SECV) from the calibration set (divided into five segments). Generally, the SECV fell sharply for the first few factors and decreased gradually for subsequent factors, which is typically indicative of a model that has properly extracted the analyte information. At some stage of the cross-validation plot, the SECV was always found to increase or remain steady. The factor before this change in trend was observed was chosen as the one that was optimal for modeling. The results are summarized in Table 1.

The same number of factors was used with both the limited and full range of data. However, the resulting SEP using the

Table 1
PLS calibration results using Raman and NIR spectroscopy

Technique used	Spectral range (cm^{-1})	SECV (%)	SEP (%)
Raman	1700–200	0.28 (3)	0.44
Raman	1700–800	0.32 (3)	0.39
NIR	10000–7500	0.30 (3)	0.36

The numbers in parentheses correspond to the number of factor used.

1700–800 cm^{-1} range was superior indicating that there was value in excluding the less reproducible spectral range. The resulting SEP (0.39%) is quite comparable to that obtained in our previous study [9] in which rotating ambroxol pellets were measured with a conventional small-spot scheme (0.35%). In other words, using the WAI scheme, a similar accuracy was achieved for comparable samples measured within their capsule shells without rotation.

For the in-depth investigation of PLS result, we have examined first three loadings (factors) in the 1700–800 cm^{-1} as shown in Fig. 6. For comparison, Raman spectrum of pure ambroxol is also displayed. The loadings were artificially offset for easy comparison. The overall features of the first and second loadings are quite close to that of ambroxol, indicating proper PLS modeling without serious interferences from other components. Although overall weights are decreased in the third loading, there are still more weights on major ambroxol bands located on 1037 and 822 cm^{-1} as in the first and second loadings. By examining three loadings, it can be concluded that the developed PLS model properly describe the variation of ambroxol concentration without overfitting.

3.3. PLS regression using NIR spectral data

The method currently used most often for non-destructive quantitative measurements in the pharmaceutical industry is Near-Infrared (NIR) spectroscopy [10,11]. As such, it is worthwhile to compare the results of the Raman measurements from this study to similar measurements done with NIR. The WAI scheme is very useful for reliable quantitative performance with Raman measurements as shown in this and other studies [4,5]. However, if the measurement accuracy in this case or other cases is not substantially superior to that of NIR spectroscopy, there would be little value in considering the implementation of a routine Raman method over one using NIR. This is due to the fact that NIR is currently more widely accepted and is cheaper to implement compared to Raman. However, Raman often offers better specificity than NIR and, thus, simpler method development and less long-term calibration maintenance.

For the purpose of this study, we decided to employ a transmission measurement scheme to collect NIR data rather than one using diffuse reflectance. This was based on several reports suggesting that the transmission mode of collection for solid pharmaceutical samples provided improved measurement accuracy relative to diffuse reflectance [12,13]. Capsule contents were transferred to 2 mm pathlength quartz cuvettes for the transmission measurements. Direct transmission through the capsules was difficult due to the analytical pathlength as well as the opacity of the capsule shell (see Fig. 1). In this case, the resulting signal-to-noise ratio for NIR through-capsule measurements would be substantially degraded.

To enhance the signal-to-noise ratio of the NIR spectra, an InAs detector was employed and data were collected with 64 co-added scans. Fig. 7 shows the NIR spectra of pure ambroxol and lactose in the 10000–7500 cm^{-1} range. The NIR features of ambroxol are relatively distinct due to the fact that it is an aromatic compound. It has a higher absorptivity compared to that of lactose. The NIR spectrum of the 12% ambroxol formulation in lactose is also shown in the figure. As expected, the resulting NIR spectral features are much broader and less characteristic compared to those in the Raman spectrum.

The whole NIR spectral range used for collection was employed for the construction of a PLS model. The results are shown in Table 1. The SECV and SEP from NIR are just slightly better or similar to those obtained from measurements employing the Raman WAI scheme. Based on *F*-test, the difference in SECV and SEP between two measurements was insignificant (95% confidence level). Hence, even though the Raman spectra were collected through the intact capsule, the measurement

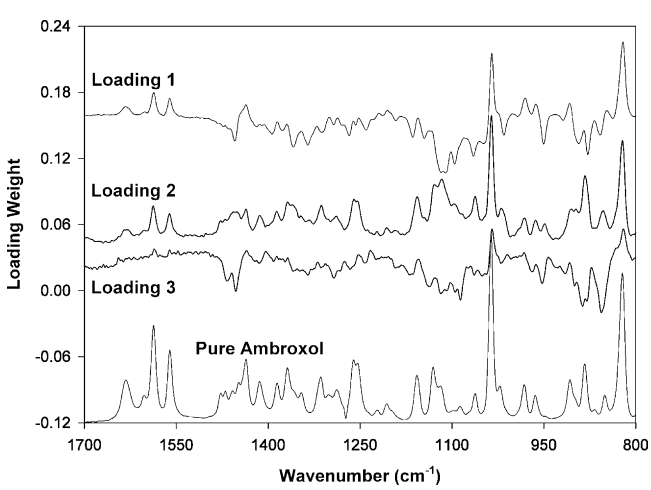

Fig. 6. First three loadings (factors) used for PLS calibration in the 1700–800 cm^{-1} range.

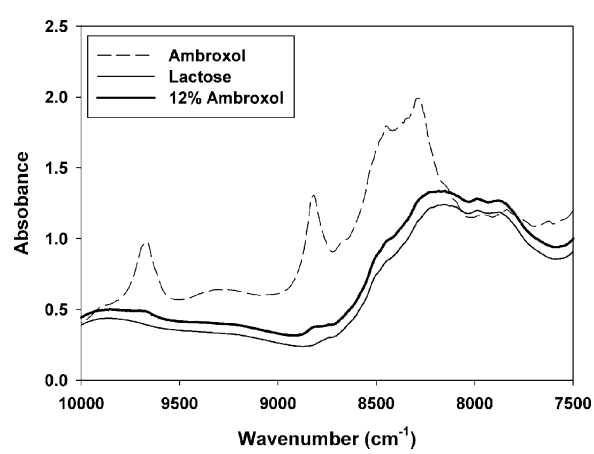

Fig. 7. NIR spectra of pure ambroxol, lactose and 12% ambroxol in the 10000–7500 cm^{-1} range.

accuracy was comparable to that of the NIR transmission measurements collected without the complicating presence of the capsule material.

4. Conclusion

The Raman WAI scheme was successfully employed to non-destructively measure ambroxol contents in formulations contained in intact capsules. The WAI scheme produced Raman spectra that were much more representative of the correct overall concentration of the ambroxol in these samples compared to a conventional scheme employing a small-spot laser. The latter tends to produce measurements that represent localized sample concentrations. This lack of overall representation introduces significant measurement errors in the determination of a bulk property such as the active ingredient level. The results of this study suggested that the interference of the capsule material on the Raman measurements was not substantial when the WAI scheme was employed. This approach should be applicable to other circumstances that require the non-destructive analysis of APIs in formulations contained in other capsule types, such as gel capsules. The results obtained in this study suggested that quantitative Raman measurements employing the WAI scheme might offer advantages in some cases relative to similar measurements using NIR spectroscopy. This is significant since NIR is currently the benchmark technique for non-destructive quantitative analyses in a PAT setting. This implies that Raman measurements using this approach should be considered when rapid, non-destructive quantitative measurements are required as part of a process analysis plan.

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